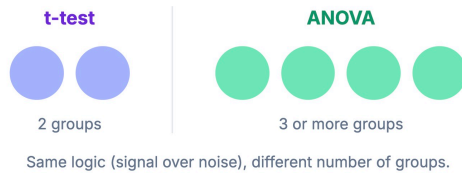


RSR Workshop 05

1. What is the main difference between a t-test and a one-way ANOVA?



a) A t-test is for two groups, ANOVA is for more groups ✓

b) A t-test is for paired data, ANOVA is for unpaired data

c) ANOVA needs normal data, but a t-test does not

2. The t-statistic is signal divided by noise. What makes it larger (and the p-value s

t-Test Formula


$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$
$$t = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$


a) A higher chosen alpha

b) A bigger gap between means, or less noise ✓

c) A bigger standard error or more noise

3. From this gender t-test, what do we conclude at alpha = 0.05?

```
data: total_scores_of_K6 by Gender
t = -2.86, df = 1597, p-value = 0.0043
95 percent confidence interval: -1.41 -0.26
mean Female = 7.9    mean Male = 8.7
```

a) Fail to reject the null; the means look equal

b) The result is invalid because t is negative

c) Reject the null; the mean scores differ ✓

4. In an Age-category ANOVA output, which value is the p-value?

4. AGE-CATEGORY ANOVA TABLE

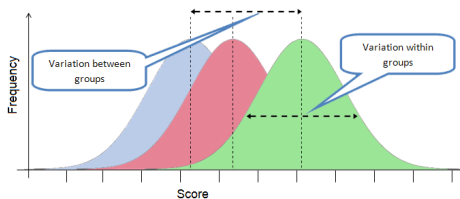
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Age category	3	612	204.0	9.8	2.1e-06
Residuals	1595	33215	20.8		

a) Df = 3

b) $\text{Pr}(>F) = 2.1\text{e-}06$ ✓

c) F value = 9.8

5. The age-category ANOVA is significant. What does that tell us?



a) Every age group differs from every other group

b) The oldest group has the highest distress

c) At least one age group differs from the rest ✓

6. What is a Type I error?

	no real effect	real effect exists
Reject null	Type I false positive	correct
Keep null	correct	Type II false negative

a) Claiming an effect that is not really there ✓

b) Failing to detect a real effect

c) Picking the wrong test for your data

7. When should you typically do a sample size/power analysis?

```
> power.t.test(delta = (3600 - 3420),
+             sd = 500,
+             power = .9
+             )
Two-sample t test power calculation
  n = 163.1179
 delta = 180
  sd = 500
sig.level = 0.05
 power = 0.9
alternative = two.sided
NOTE: n is number in *each* group
```

a) Before a study ✓

b) During the recruitment phase of a study

c) After receiving study results, to determine how to report.